The Paradox of Authorial Identity: Measuring Style Decay and Semantic Retention Under Iterative LLM Paraphrasing

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1 PROBLEM DESCRIPTION

A machine paraphrase preserves *what* a text says while rewriting *how* it says it. Applied repeatedly, this poses the Ship of Theseus question for authorship: after a document has been paraphrased three times by a large language model, whose text is it? Authorship attribution, machine-generated text (MGT) detection, plagiarism analysis, and forensic provenance all assume a stable stylistic signal survives in a document; paraphrasing cheaply defeats that assumption.

We therefore measure identity decay quantitatively rather than as a single pass/fail outcome, tracking generation by generation which linguistic properties are destroyed first and which persist. Three questions organize the work: **(RQ1)** which features decay before semantic content does; **(RQ2)** at which generation attribution becomes unreliable; and **(RQ3)** whether the paraphraser leaves a recoverable fingerprint, so that authorial identity is not merely erased but *replaced*. Our contribution is a decay curve separating human-author signal loss from model-fingerprint gain.

2 BACKGROUND

Iterative paraphrasing and authorship. In the work done by Tripto et al. [4] at ACL 2024, the authors ask whether a text keeps its original authorship after repeated paraphrasing. They apply four paraphrasers (ChatGPT, PaLM2, Pegasus, DIPPER) iteratively over seven datasets, re-running attribution and MGT detectors at each generation. Attribution degrades, but they report only aggregate accuracy, never isolating which features decay.

Paraphrase as an evasion attack. In the work done by Krishna et al. [1] at NeurIPS 2023, the concern is detector robustness to evasion. They train DIPPER, an 11B-parameter discourse-level paraphraser with controllable lexical diversity and reordering, and propose a retrieval defense. One pass drives DetectGPT and GPTZero toward chance, but the defense presumes the generating API logs its outputs.

Zero-shot MGT detection. In the work done by Mitchell et al. [2] at ICML 2023, the goal is detecting generated text without labeled data. DetectGPT compares a candidate's log-likelihood against those of small perturbations of itself, exploiting negative curvature in the probability surface. It beats prior zero-shot baselines but needs white-box likelihoods and collapses under paraphrase attack.

Attribution across neural generators. In the work done by Uchendu et al. [5] at EMNLP 2020, attribution is extended from humans to neural generators and cast as three tasks. They benchmark stylometric, deep-learning, and transformer classifiers over nineteen generators. Generators stay separable, but only single-pass text is tested, leaving open whether signatures survive rewriting.

Content-independent author representations. In the work done by Rivera-Soto et al. [3] at EMNLP 2021, the question is whether one authorship representation transfers across domains. They train a Transformer encoder contrastively on Reddit, Amazon, and fanfiction so same-author documents embed together. The embeddings transfer well but stay sensitive to topic overlap—a style/content confound our fixed-source design removes.

Measuring semantic retention. In the work done by Zhang et al. [6] at ICLR 2020, the problem is evaluating text beyond n-gram overlap. BERTScore greedily matches candidate and reference tokens using contextual embeddings. It tracks human adequacy judgments far better than BLEU but is blind to style, so we pair it with explicit stylometry.

3 DATASET AND REPOSITORY

We use the Ship of Theseus paraphrased corpus [4]: seven source datasets × four paraphrasers (ChatGPT, PaLM2, Pegasus, DIPPER), each sample linked to its originating document across generations T0 (human original) through T3. We start with `xsum` and `yelp`, extending to the rest for cross-domain validation. *Status:* selected; file access and schema verification in progress. Repo: https://github.com/gajawp/ShipOfThesus.

4 PROJECT PLAN AND MILESTONES

Wk	Activity and deliverable
3	Acquire corpus; verify schema; parse T0–T3 lineage (`xsum`, `yelp`).
4	Preprocessing pipeline; leakage-free document-level splits.
5	Semantic baselines (BLEU, ROUGE-L, BERTScore, SBERT). **M1: data pipeline.**
6	Lexical features: type–token ratio, function words, punctuation, length.
7	Syntactic/readability features: POS n-grams, dependency patterns, Flesch–Kincaid.
8	Decay curves; style-vs.-semantics tests. **M2: content audit (RQ1).**
9	Attribution baselines on T0 only: stylometry + SVM, fine-tuned RoBERTa.
10	Evaluate T0-trained models on T1–T3; per-generation accuracy decay.
11	Locate "point of no return"; error analysis. **M3: breakdown (RQ2).**
12	Four-way paraphraser classifier, held-out data. **M4: fingerprints (RQ3).**
13	Identity-trajectory visualizations, Streamlit dashboard, ablations.
14	Final ACL-format paper; repo docs; presentation. **M5: deliverables.**

Gates throughout: no cross-split document leakage; macro-F1 per classifier; fingerprints validated on unseen data.

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